Can You Beat Randomness?

An Evaluation of Football Betting Market Efficiency,
Machine Learning Calibration, and Gambler's Ruin

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https://github.com/meh6di/can-you-beat-randomness

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Abstract

This report studies whether statistical models can generate a durable profit against the pricing of European football betting markets. Using three seasons of match and odds data from the top five leagues (Premier League, La Liga, Serie A, Bundesliga, Ligue 1), we train multinomial logistic regression and XGBoost models (raw and Platt-calibrated) on 2021–2023 and evaluate them out-of-sample on 2023–2024. A value-betting strategy bets whenever the model probability exceeds the overround-free market probability by a fixed margin. A blind random bettor loses roughly 7% of stakes to the bookmaker margin. The best model reaches an aggregate ROI of -1.03% over 752 bets, which, without a formal significance test and given the small sample, cannot be distinguished from either zero or the random-bettor baseline. Strong results in single leagues rest on small samples and are consistent with variance. Even with an ROI close to zero, the equity curve suffers drawdowns larger than a typical retail bankroll. The study is for educational purposes only and is not betting advice.

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1 Introduction

Betting advertising is ubiquitous around football. This project asks a simple question: *can algorithmic models consistently exploit inefficiencies in European football betting markets, or does the bookmaker margin guarantee long-term decline?*

The contribution is a fully reproducible pipeline that (i) downloads public data, (ii) builds leakage-free features, (iii) benchmarks models against naive agents under identical rules, and (iv) quantifies the risk of ruin through drawdown and Monte Carlo analysis.

Disclaimer. This work is solely for educational, academic and research purposes in quantitative data analysis and machine learning. It is not financial or gambling advice.

2 Data

2.1 Source and scope

Data come from `football-data.co.uk` (`fetch_data.py`), one CSV per league and season:

- **Leagues:** Premier League (E0), La Liga (SP1), Serie A (I1), Bundesliga (D1), Ligue 1 (F1).
- **Seasons:** 2021–2022, 2022–2023, 2023–2024.

2.2 Cleaning

`clean_data.py` keeps only the columns needed for modelling and benchmarking: result and goals (`FTHG, FTAG, FTR`), shots and shots on target, corners, fouls, cards, market-average odds (`AvgH, AvgD, AvgA`) and Pinnacle odds (`PSH, PSD, PSA`). Dates are parsed (day-first) and sorted chronologically, and rows missing the result or average odds are dropped. Pinnacle odds are retained but not yet used by the models (see Section 8).

3 Feature Engineering and Leakage Prevention

All features are built per league-season file in `engineer_features.py`.

3.1 Overround-free market probabilities

For decimal odds o_H, o_D, o_A the bookmaker's overround is

$$\Omega = \frac{1}{o_H} + \frac{1}{o_D} + \frac{1}{o_A} > 1, \tag{1}$$

and the market-implied fair probability of outcome $k \in \{H, D, A\}$ is

$$p_k^{\text{mkt}} = \frac{1/o_k}{\Omega}. \tag{2}$$

3.2 Team-level rolling form

Each match is reshaped into two team-match rows (home and away perspective). For every team, rolling 5-match means are computed on points, goals for/against, shots for/against, shots on target (SoT) for/against, corners for/against, fouls and cards. Look-ahead bias is avoided by applying `shift(1)` before the rolling window, so a match's features only use earlier matches. The window uses `min_periods=1`, so early-season rows rely on few games, and the first match of each team in a file (no history) is dropped.

3.3 Derived indicators

- **Shot accuracy:** rolling SoT divided by rolling shots.
- **Conversion:** rolling goals divided by rolling SoT.
- **Rest days:** days since the team's previous match, capped at 14 (missing set to 7), as a fatigue/congestion proxy.
- **Differentials:** home minus away for points, SoT, corners and rest days.

3.4 Model inputs

The 19 model features are the home/away rolling points, goals, SoT, corners, accuracy and conversion, the four differentials, and the three market probabilities. Including p^{mkt} as input means the models start from the market's own information and must add value on top of it.

```
fetch_data.py -> data_raw/ -> clean_data.py -> data_clean/
   -> engineer_features.py -> data_features/
   -> backtest.py / backtest_advanced.py / deep_analysis.py
      / plot_gambling_risks.py
```

Figure 1: Pipeline overview.

4 Methodology

4.1 Train/test protocol

For each league a separate model is trained on 2021–2022 and 2022–2023 and tested on 2023–2024. The split is strictly chronological, so no future information enters training.

4.2 Models

- **Multinomial logistic regression** (`max_iter=1000`).
- **XGBoost** with 75 trees, depth 3, learning rate 0.05, subsample 0.8, column subsample 0.8, multi-class log loss.
- **Platt-calibrated XGBoost:** the same XGBoost wrapped in `CalibratedClassifierCV` (sigmoid, 3-fold) on the training data.

4.3 Betting agents and profit

Each bet stakes one unit at the market-average odds o. The profit is $o - 1$ if the bet wins and -1 otherwise; ROI is total profit divided by number of bets. Agents:

1. **Random:** picks H, D or A uniformly on every match.
2. **Favorite:** always bets the lowest-odds outcome.
3. **Model (value betting):** bets outcome k when the edge

$$e_k = \hat{p}_k - p_k^{\text{mkt}} \tag{3}$$

exceeds a buffer δ.

In the first backtest $\delta = 3\%$ and several outcomes per match may be bet. In the advanced backtest $\delta = 6\%$ and at most one bet is placed per match, on the outcome with the largest edge.

4.4 Risk analysis

`plot_gambling_risks.py` merges the five leagues' test predictions chronologically, as a real bettor would experience them, and computes the cumulative profit path of the logistic model. It compares this with 100 Monte Carlo random bettors placing unit bets on the same matches, and computes the drawdown (distance from the running peak) with a hypothetical 20-unit bankroll bust threshold.

5 Results

5.1 Benchmark backtest (3% buffer)

Agent	Bets	Profit (units)	ROI
Random	1,704	−117.74	−6.91%
Favorite	1,704	+2.40	+0.14%
Model (LogReg)	1,502	−6.43	−0.43%

Table 1: Aggregate results over the five leagues, 2023–2024.

The random agent's loss is consistent with the bookmaker's margin acting as a "gravity well". The model recovers most of that loss but does not beat zero.

5.2 Advanced backtest (6% buffer, one bet per match)

Competition	LogReg bets	LogReg ROI	XGB bets	XGB ROI
Premier League	155	+4.00%	228	−17.16%
La Liga	150	+3.18%	225	−6.24%
Serie A	126	+29.90%	188	−8.13%
Bundesliga	203	−28.17%	208	−3.18%
Ligue 1	118	+0.69%	159	+7.23%
Overall	**752**	**−1.03%**	**1,008**	**−6.31%**

Table 2: Per-league results; XGBoost is the uncalibrated model.

Logistic regression outperforms the uncalibrated trees overall. A plausible explanation is that tree ensembles produce step-wise, overconfident probabilities that generate many spurious "edges", while the smooth log-odds of logistic regression stays closer to the market. The extreme league results (+29.90% in Serie A, −28.17% in Bundesliga) come from 120–200 bets each, a sign of high variance rather than stable skill.

Bundesliga. The loss is notable. Out-of-sample regime shifts, such as Bayer Leverkusen's undefeated 2023–2024 season, break models relying on a two-year history, illustrating the non-stationarity of sports dynamics.

5.3 Calibration analysis

`deep_analysis.py` evaluates Platt-calibrated XGBoost against logistic regression with a per-outcome breakdown (bets, win rate, profit, ROI for H/D/A) and equity curves. No calibrated-XGBoost ROI is included in the table above, which reports the raw model.

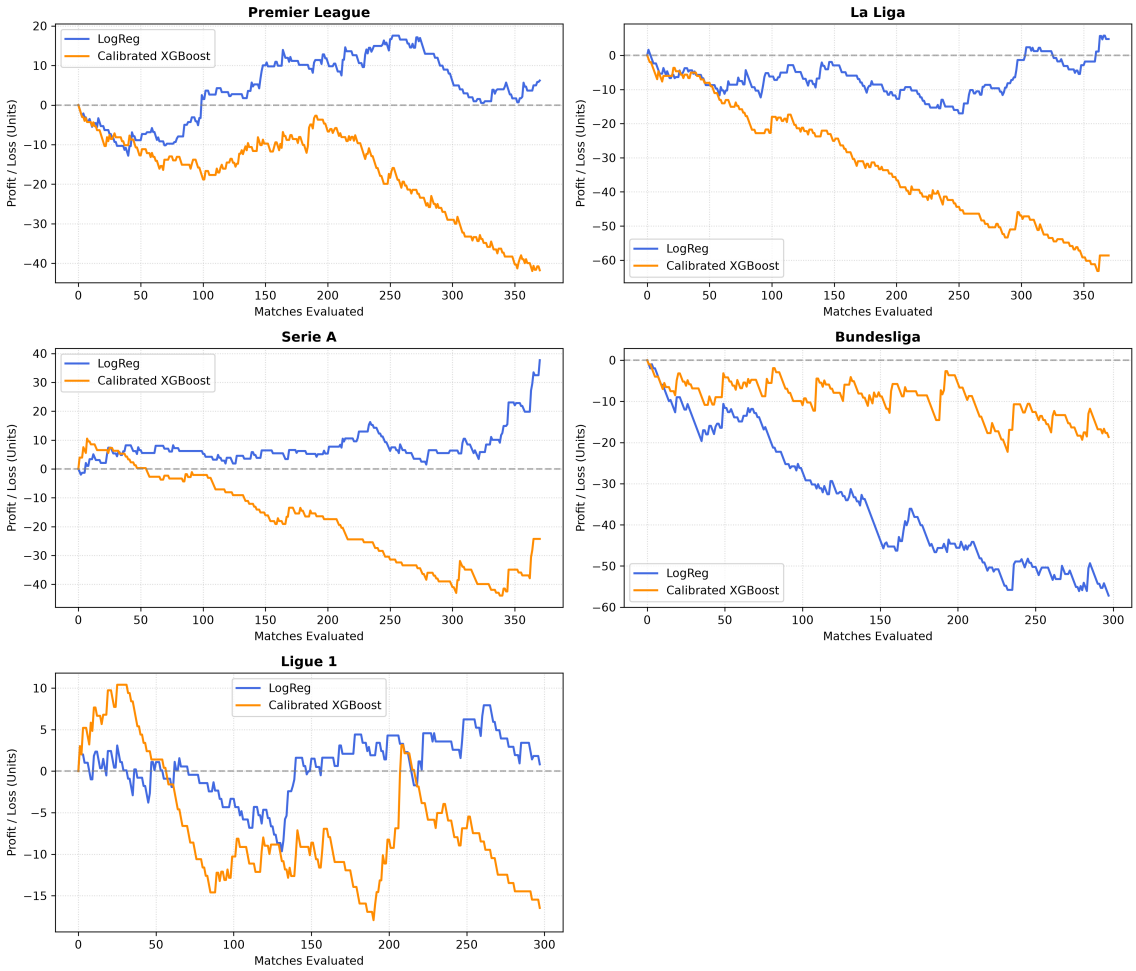


Figure 2: Cumulative profit per league for logistic regression and calibrated XGBoost.

5.4 Risk of ruin

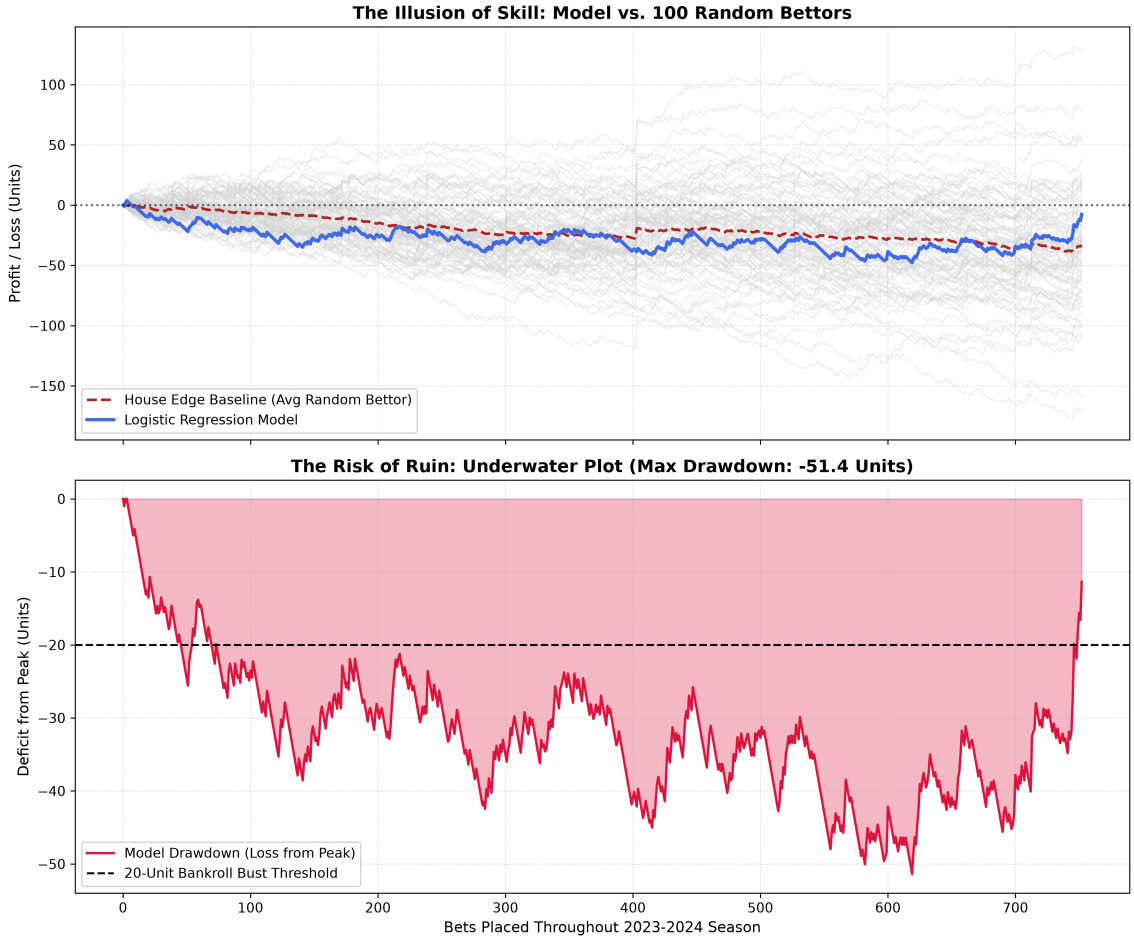


Figure 3: Top: logistic model versus 100 random bettors. Bottom: underwater (drawdown) plot with the 20-unit bust threshold.

- **Illusion of skill:** the model's equity path lies within the cloud of random bettors, so upward phases are statistically consistent with ordinary positive variance.
- **Drawdown exceeds capital:** although the season ends with an ROI close to zero (-1.03%), the maximum peak-to-trough drawdown reaches -51.4 units, far beyond the 20-unit threshold, which is crossed within the first 100 bets. A retail bettor with such a bankroll would be wiped out before recovery.

6 Discussion and Limitations

- **Sample size:** one test season; per-league results use 118–228 bets, too few to separate skill from luck. No significance test is reported, so the aggregate ROI of -1.03% over 752 bets is not statistically distinguishable from zero or from the -7% random baseline.
- **Threshold selection:** the buffer was changed from 3% to 6% between scripts. If chosen after viewing test results, this introduces mild selection bias, and thresholds should be tuned on a validation season.
- **Odds realism:** profits use market-average odds. Real bettors face line movement, limits and account restrictions, usually worsening outcomes.

- **Staking:** flat one-unit stakes only; Kelly or fractional staking is not explored.
- **Features:** short 5-match windows with `min_periods=1` are noisy early in the season; no team strength rating or cross-season carry-over is used.
- **Comparison with random bettors:** the Monte Carlo bettors bet on the same matches as the model but choose outcomes randomly; this isolates pick quality, not match selection.

7 Conclusion

After the bookmaker's margin, none of the models produced returns distinguishable from luck over the 2023–2024 season. A random bettor loses about 7%, smoother models fare better than uncalibrated trees, and apparent per-league successes rest on small samples. Even an aggregate ROI close to zero carried drawdowns large enough to ruin a typical bankroll. In this study, modelling skill did not translate into a reliable betting edge.

8 Future Work

- Dixon–Coles Poisson goal-expectancy models.
- Closing line value (CLV) measured against Pinnacle odds, a lower-variance test of edge than realised profit.
- More seasons, rolling retraining, and formal significance testing.

9 Reproducibility

Requirements: Python 3.9+, `pandas`, `numpy`, `scikit-learn`, `xgboost`, `matplotlib`. Run in order:

```
python fetch_data.py
python clean_data.py
python engineer_features.py
python backtest.py
python backtest_advanced.py
python deep_analysis.py
python plot_gambling_risks.py
```

Acknowledgments

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